# Deep Learning (Spring 2026)
# Assignment 2

**Submission Instructions.** You are **required** to download and use the assignment solution template from the web learning portal. Failure to use the provided template may result in a lower grade. Your final submission must be a `.zip` archive containing both a `main.tex` file and a compiled `main.pdf` file.

**Grading Policy.** Please note that your grade will primarily be evaluated based on the contents of your `main.tex` file.

## 1 True or False

**Problem 1. (10 pts)** Generative models can be used to both classify and generate images.

**Problem 2. (10 pts)** We can train an energy-based model without knowing the explicit density function (or normalizing factor $Z$).

**Problem 3. (10 pts)** Stochastic Gradient MCMC is designed to solve the optimization problem $\arg\max_\theta P(\theta|X)$, where $\theta$ is the collection of parameters and $X$ represents data.

## 2 Q&A

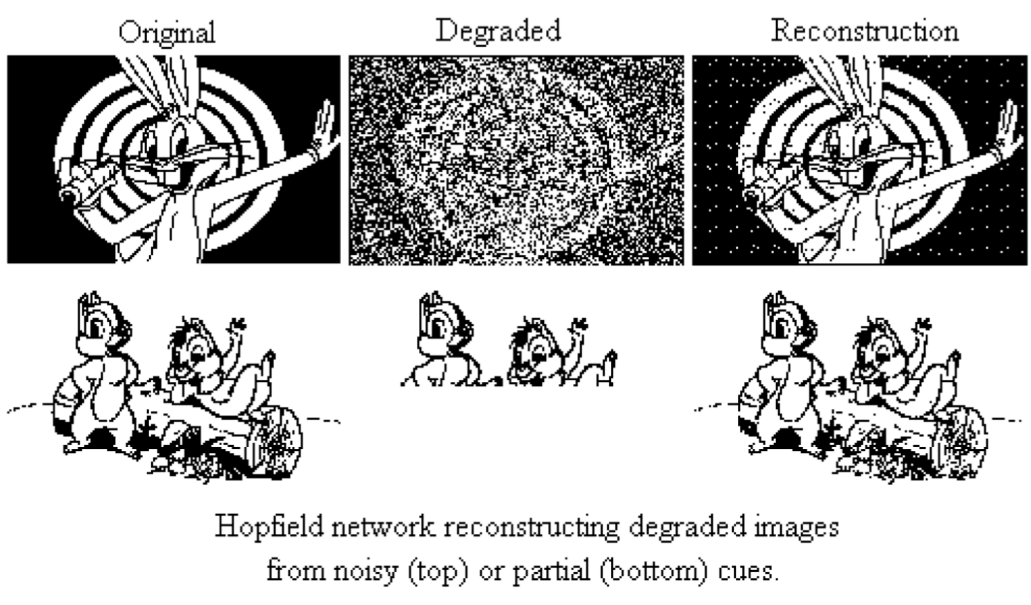


Figure 1: Noisy image (top row) and masked image (bottom row).

**Problem 4. (30 pts)** (Hopfield Network) Answer the following questions about the Hopfield network.

1. Figure 1 shows two types of degraded images: noisy image and masked image. Design an appropriate process to retrieve stored patterns using the Hopfield network for each case respectively. (Hint: The unmasked part of a masked image is the same as ground truth. By contrast, most pixels of a noisy image are different from the ground truth.)

2. (Redundancy) Although a Hopfield network explicitly stores only $N$ patterns, redundancy in its state space allows it to represent many more configurations. Suppose the Hebbian learning rule is given by $W = \frac{1}{N}\sum_p y_p y_p^T$. We want to use the Hebbian learning rule to construct a Hopfield network. Prove that with $N$ orthogonal patterns $y_p$ ($y_p$ is a $N$-dim vector) for $p = 1, \ldots, N$, the Hopfield network can memorize all $2^N$ patterns, in the sense that each of these patterns corresponds to a local minimum of the network's energy function.

**Problem 5. (30 pts)** (Boltzman Machine) Consider a fully connected Boltzman machine. We remark the visible units as $v$, the hidden units as $h$, and all units $y = (v, h)$. The joint probability of $v$ and $h$ is given by

$$P(v, h) = \frac{\exp(y^T W y)}{\sum_{y'} \exp(y'^T W y')}.$$

And the marginal probability of $v$ is given by

$$P(v) = \sum_h P(v, h).$$

We aim to maximize the log-likelihood, and the loss is given by

$$L(W) = -\frac{1}{|\mathcal{P}|}\sum_{v \in \mathcal{P}} \log P(v).$$

Prove that the gradient of Eq. (3) has the following form:

$$\nabla_W L(W) = -\frac{1}{|\mathcal{P}|}\sum_{v \in \mathcal{P}} \left(\mathbb{E}_{h|v}[yy^T] - \mathbb{E}_{y'}[y'y'^T]\right).$$

**Problem 6. (30 pts)** (Gaussian RBM) Consider a restricted Boltzman machine with a single hidden layer and the following energy function $E_{W,b} : \mathbb{R}^{N_h + N_v} \to \mathbb{R}$:

$$E_{W,b}(v, h) = \frac{1}{2}(v - b)^T (v - b) - v^T W h$$

where $W$, $b$ are trainable parameters, $v$ is visible continuous-value units (i.e., $v \in \mathbb{R}^{N_v}$), and $h$ is hidden discrete-value units (i.e., $h \in \{-1, 1\}^{N_h}$).

1. Derive the conditional distribution $P(v|h)$.

2. Derive the gradient of $b$ if we train this model based on the maximum log-likelihood principle. (Hint: Your answer should contain the form of an expectation.)

**Problem 7. (30 pts)** (Undirected Probabilistic Model) A graphical model or probabilistic graphical model or structured probabilistic model is a probabilistic model for which a graph expresses the conditional dependence structure between random variables. In an undirected graphical model, an edge implies dependence between the corresponding random variables. Figure 2 shows an example, where the joint probability distribution can be factorized as

$$P(A, B, C, D, E, F) = \frac{1}{Z} f_{AD}(A, D) f_{AC}(A, C) f_{AE}(A, E) f_{BC}(B, C) f_{EF}(E, F)$$

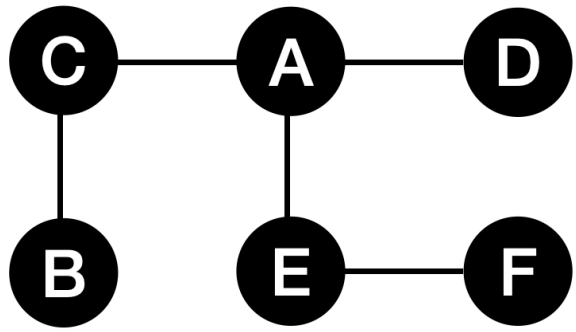


Figure 2: An undirected graphical model example (variables $A$–$F$).

for some non-negative functions $f_{AB}, f_{AC}, f_{AD}, f_{AE}, f_{BC}$, and $f_{EF}$, and a normalizing factor $Z$ (also called partition function).

1. Are $D$ and $F$ independent?
2. Write down the unnormalized conditional distribution $P(B, E|A)$. "unnormalized" means you can omit the normalizing factor. Are $B$ and $E$ independent given $A$?
3. We model $P(A, B, C, D, E, F)$ as a Boltzman distribution

$$P(A, B, C, D, E, F) \propto \exp(-E(A, B, C, D, E, F))$$

where $E$ is the energy function. Show that the energy function can be expressed by the following factorization:

$$E(A, B, C, D, E, F) = E_{AC}(A, C) + E_{AD}(A, D) + E_{AE}(A, E) + E_{BC}(B, C) + E_{EF}(E, F).$$

**Problem 8. (30 pts)** (Importance Sampling) $x$ is a random variable. Given target distribution $p(x)$ and target random variable $y = f(x)$, importance sampling gives an estimator of $\mathbb{E}[y]$ from a proposal distribution $q(x)$:

$$\mathbb{E}_{x\sim p}[f(x)] = \mathbb{E}_{x\sim q}\left[\frac{p(x)}{q(x)} f(x)\right] \approx \frac{1}{N} \sum_{x\sim q(\cdot)} \frac{p(x)}{q(x)} f(x).$$

Prove that when $q$ has the following form,

$$q^\star(x) \propto p(x)|f(x)|$$

the variance of this estimator can be minimized.

**Problem 9. (30 pts)** (Markov Chain Monte Carlo)

1. Prove random-walk Metropolis-Hasting sampling (i.e., $s' \leftarrow s +$ Gaussian noise) is a valid MCMC algorithm, i.e., it constructs a Markov chain which is ergodic and satisfies the detailed balance property.
2. Prove that Gibbs sampling is a special case of Metropolis-Hasting sampling, and that the acceptance rate of Gibbs sampling (i.e., $\alpha(s \to s')$) is 1. Here we consider the following 2-step Gibbs proposal: (1) randomly sample a coordinate index $i$; (2) sample coordinate $s_i$ from the coordinate proposal $q(s_i \to s_i') = p(s_i'|s_{j\neq i})$.
3. (Optional Question) In fact, Gibbs sampling is typically implemented in a cyclic fashion, i.e., running posterior sampling in a fixed order over all the dimensions. Prove that cyclic Gibbs sampling yields the same stationary distribution as random-order Gibbs sampling in the above question, as long as the Markov chain can access all states under the fixed ordering.